

GRADE 6 MATHEMATICS

Classroom Observation Guide

Grade 6 Mathematics — Day Seven: "Area of Compound Rectangles"

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Prepared for visiting guests, administrators, and instructional observers.

Purpose of This Guide

Thank you for observing our Grade 6 mathematics classroom today. Day Six ended with students splitting an L-shaped floor into two rectangles and adding the pieces — their first look at a compound figure. Day Seven keeps that same L-shaped floor but hands students a second, equally valid tool for finding its area, gives them a real choice between the two, and then raises the stakes with a T-shape where the choice is no longer obvious. This guide explains how today's lesson was built, the standards it addresses, and — because this cycle's classroom outcome is directly relevant to scheduling and resourcing decisions — the observation recorded after teaching it, together with a specific support recommendation on page 2.

The Design Decision: Naming Both Moves on Purpose

Day Six's observer note was specific: most students could split an L-shaped floor into two rectangles when the split line was already drawn for them, but several froze the moment they had to decide *where* to draw it themselves — and almost no one thought to try surrounding the shape in one big rectangle and subtracting instead. That note shaped Day Seven's entire structure. Rather than continuing to treat "split it and add" as the only correct move, Day Seven opens by naming **two** legitimate strategies — Decompose-and-Add, Enclose-and-Subtract — and showing both worked side by side on the identical floor before asking students to choose. A choice a student cannot name is not really a choice; Model It, Two Ways exists specifically to make both options visible and nameable before independent work begins.

From One Tool to Two

The day's opening routine, Warm-Up Recitation, rehearses both directions before any drawing happens. Four of six cards restate yesterday's add-the-pieces language ("a floor is two rectangles, 8×3 and 4×2 ..."); two flip the same idea into an enclose-and-subtract problem ("a 10×6 rectangle is missing a 3×2 corner..."). Model It, Two Ways then draws the exact same L-shaped floor twice — once decomposed and added, once enclosed and subtracted — so students can confirm for themselves that two different-looking paths land on the same number. The T-shape that follows is where the lesson stops being a demonstration and becomes a genuine decision: a T has two notches, not one, so enclosing it means subtracting twice instead of once. Most students find adding the two pieces faster for that shape — the first time in the unit a student's choice of method actually matters.

What We Observed

Today's task asked more of students than any prior day this unit: hold two named strategies in mind, apply either one correctly to an L-shaped floor, then judge which strategy actually works better on a shape — the T — where the "easy" choice (enclose and subtract) quietly gets harder instead of easier. This is a genuinely high-level Common Core task for a sixth grader, closer to the reasoning MP.1 describes ("make sense of problems and choose a strategy") than most grade-6 area work asks for.

A meaningful part of the class adapted to that demand well: they used Model It, Two Ways to check their own work, reasoned correctly about why the T-shape favors addition, and moved through Practice: Mixed Shapes without needing redo help. Another part of the class could not adapt to the assignment at this level of difficulty — not during the warm-up recitation, and not once the independent Try It and Practice screens removed the worked example they could copy.

Why This Looks Like a Fluency Gap, Not a Decomposition Gap

The students who struggled today were generally not confused about *where* to split the shape or *which* rectangle to enclose — the spatial part of the task that 3.MD.C.7d and today's Design Decision (page 1) were built to repair. Their breakdown came one step later: once a shape was correctly decomposed or enclosed, they could not reliably execute the multiplication and division sentences that decomposition produces. That distinction matters, because it points to a different, more foundational gap than today's lesson targets. It is consistent with what Day Six's Find the Missing Side diagnostic already flagged for a number of these same students, and with the multiplication-and-division fact fluency called for by the grade-3 standard 3.OA.C.7 — fluency this group has now shown, across several days of formative evidence, that it does not yet have.

Recommendation

Move students who could not engage with today's task into a dedicated support room for focused multiplication-and-division instruction, rather than routing them only to the in-class Engage station. The in-class Engage rotation is built to repair a same-day, same-lesson misconception (today: which method to choose); it is not built to rebuild a missing grade-3 operations foundation, which needs more direct instructional time than a station rotation inside a 55-minute grade-6 period can give it. A separate support-room block, run in parallel with continued grade-6 content exposure, targets the actual skill gap directly.

How This Fits Our Tiered Support Model

Day Six's guide cited TNTP's *The Opportunity Myth* (2018) finding that students make more progress on grade-appropriate assignments than on remedial work assigned in place of them — and that principle still holds here. This recommendation does not pull identified students off grade-6 content; it adds a targeted, time-boxed Tier 3 block for a specific, repeatedly diagnosed prerequisite gap, alongside their continued participation in Day Eight and beyond. That combination — grade-level content as the default, with a short, explicit block for a named prerequisite skill — is the model described in the Response to Intervention literature (Fuchs & Fuchs, 2006; Gersten et al., 2009) and matches the National Mathematics Advisory Panel's (2008) conclusion that fluency with whole-number operations is a critical foundation later math work is built on, not a nice-to-have.

Before Day Eight

- Confirm the specific student list from today's Model It, Practice: Mixed Shapes, and Exit Ticket results, rather than relying on this note alone.
- Coordinate a support-room block and staffing with the intervention team; today's diagnostic points specifically at multiplication/division fact fluency and the related unknown-factor (division) reasoning from 3.OA.B.6, not at area or geometry content.
- Set a simple entry/exit check (e.g., a short fact-fluency probe) so placement is revisited weekly against new formative data, rather than becoming a standing assignment.
- Carry this note forward as Day Eight's opening observer note, the same way Day Six's note shaped today's Design Decision on page 1.

Lesson at a Glance (55 Minutes)

Time	Segment	What's Happening
0–3 min	Welcome Back	Students sign in on the lesson app. Day Six's observer note is read aloud, and the class hears today's goal: two named tools for one compound floor, and a real choice on a trickier T-shape.
3–8 min	Warm-Up Recitation	Six cards, read one at a time — four "add the pieces," two "subtract the missing piece," plus two reverse cards that work backward from a known total. The class chorally finishes each card.
8–16 min	Try It: The New Floor Plan	Sliders build an L-shaped floor live. Students fill in Piece A, Piece B, and the total before checking — the first graded attempt at decompose-and-add on a shape they did not see pre-split.
16–26 min	Model It, Two Ways	The identical L-shaped floor is solved twice, side by side — once decomposed and added, once enclosed and subtracted — so students confirm both methods agree before choosing either one on their own.
26–34 min	Discuss It & Connect It	A T-shape is introduced. Because a T has two notches instead of one, enclosing it means subtracting twice — the class discusses why most people find adding the two pieces faster for this particular shape.
34–42 min	Practice: Mixed Shapes	Randomized L-shape and T-shape problems, solved by whichever method the student chooses, with redo help after a second miss.
42–48 min	Engage / Explore / Enrich	Three boards, grouped by what the teacher observed during Model It and Practice. Enrich previews a diagonally-cut corner — the bridge into Area of Triangles.
48–53 min	Apply It: Exit Ticket	Silent, three required problems plus an Enrich-only fourth problem with two separate cut-outs. Students must show which method — add or subtract — they used for each.
53–55 min	Journal & Wrap	Students sketch their own three-rectangle compound figure worth at least 50 square units, then trade with a partner to solve it a different way and check that both totals match.

Why Two Named Methods, Not One?

Teaching "the missing-side trick" or "the split-it trick" as a single default procedure would have let today's lesson pass quietly for most students — but it would have hidden exactly the judgment Day Six's observer note flagged as missing: the ability to look at a shape and decide how to attack it, rather than reaching for the one method that was taught. Naming both methods, proving they agree on the same shape, and then handing students a shape (the T) where the two methods are no longer equally efficient is what turns "decompose a compound figure" from a memorized step into an actual strategic choice — the heart of MP.1.

Today's Classwork Worksheet: "Compound Area — Two Ways to Find a Compound Figure's Area"

DEC_MPO_E → DECOMPOSE

Each student receives a one-page classwork worksheet (CCSS 6.G.A.1, 3.MD.C.7d, MP.1, MP.7, and the ELA standard L.6.4). Section A, Guess the Word, has students fill in two missing letters to spell DECOMPOSE, the same word-then-model design Day Six used for INVERSE. Section B, Meet the Compound Figure Family, uses a word bank (compound figure, decompose, enclose, subtract) to define both methods in the exact terms the app uses. Section C, Try It, walks students through a worked add-the-pieces example ($8 \times 3 + 4 \times 2 = 32$ sq ft) and then asks them to solve and draw their own enclose-and-subtract problem — a $10 \text{ ft} \times 7 \text{ ft}$ rectangle missing a $4 \text{ ft} \times 3 \text{ ft}$ corner ($70 - 12 = 58$ sq ft). A closing Explore/Summary panel previews next: corners that are cut on a diagonal instead of straight across become triangles, not rectangles — the on-ramp into the unit's next topic, Area of Triangles.

The Engage / Explore / Enrich Boards

Board	Who	Task	IXL Skill
Engage (with teacher)	Still unsure where to draw the split, or which method to reach for	Physically cut a paper L-shape into two rectangles and label each piece; rebuild the same L from two pre-cut rectangles; worksheet Part 1 only (add-the-pieces problems)	In-class only — no IXL assignment; see this guide's page 2 recommendation for students who need more than this station can provide.
Explore (independent)	Comfortable adding pieces, ready to try subtracting	Worksheet Parts 1 and 2 (add <i>and</i> subtract problems); solve one L-shape problem both ways and confirm the totals match	GG.11 – Area of compound figures
Enrich (independent, previews parallelograms)	Fluent with both methods, finished early	A 12 ft × 9 ft room with a 4 ft × 3 ft <i>triangular</i> corner cut off — can subtract still work, and what changes about finding the cut piece's area? Then design an original 3-piece compound figure for a partner to solve.	GG.12 – Area of compound figures with triangles

IXL codes are from IXL's published Grade 6 skill list under Perimeter and Area (GG).

Common Core & Michigan Standards Addressed

Michigan's K–12 Standards for Mathematics adopt the Common Core State Standards; the codes below are identical in both documents.

Grade-Level Standard (Grade 6)

6.G.A.1: Find the area of right triangles, other triangles, special quadrilaterals, and polygons by composing into rectangles or decomposing into triangles and other shapes; apply these techniques to real-world problems. Today's entire lesson — decompose-and-add, enclose-and-subtract, and the T-shape choice between them — is this standard taught as two explicit, comparable strategies rather than one.

Foundational Standard Being Extended

3.MD.C.7d: Recognize area as additive; find the area of rectilinear figures by decomposing them into non-overlapping rectangles and adding the areas of the non-overlapping parts, applying this technique to solve real-world problems. Day Seven is this third-grade "repair" standard applied at grade-6 depth, now with a second, equally valid inverse strategy (enclose-and-subtract) added on top of it.

ELA/Literacy Standard Addressed

L.6.4: Determine or clarify the meaning of unknown and multiple-meaning words and phrases, choosing flexibly from a range of strategies. Section A of the classwork worksheet has students infer DECOMPOSE from its letter pattern and context before applying it to today's task.

Standards for Mathematical Practice

MP.1: Make sense of problems and persevere in solving them — specifically, choosing which of two valid strategies fits a given shape, rather than applying one memorized procedure automatically.

MP.7: Look for and make use of structure — recognizing that a T-shape's two notches, not one, is exactly what makes enclose-and-subtract a worse choice there than on a simple L.

Pedagogical & Cognitive Research Foundations

Concrete–Representational–Abstract (CRA)

Students physically cut a paper L-shape into rectangles (concrete), draw and label the same shape split two different ways (representational), and only then write and compare $A = a \times h + b \times h$ against $A = \text{big} - \text{cut}$ (abstract). Witzel, Riccomini, and Schneider (2008) found this sequence particularly effective for secondary students with gaps in prior mathematics — a description that, as today's observation on page 2 confirms, still fits a meaningful portion of this class.

Comparing Solution Methods Builds Flexibility

Model It, Two Ways deliberately puts decompose-and-add and enclose-and-subtract side by side on one identical shape rather than teaching them on separate days. Rittle-Johnson and Star (2007) found that having students directly compare multiple correct solution methods on the same problem — rather than studying each method in isolation — produces stronger conceptual and procedural understanding than either method alone. The T-shape discussion that follows depends on that comparison already being available to students.

Retrieval Practice Through Choral Recitation

The Warm-Up Recitation is brief, spoken, and repeated in some form daily. Roediger and Karpicke (2006) showed that brief, repeated retrieval produces more durable learning than longer, less frequent review — the same rationale behind alternating add and subtract cards aloud rather than assigning them as a written warm-up.

Just-in-Time Remediation, and Its Limit

Day Six's guide cited TNTP's *The Opportunity Myth* (2018) finding that students assigned grade-level work make more progress than peers assigned remedial work in its place — the reasoning behind repairing square-vs-rectangle confusion inside Day Six's own lesson rather than pulling students out for a separate reteach. Today's observation (page 2) shows where that principle reaches its limit: a same-day misconception can be repaired inside the lesson, but a standing, multi-day fluency gap in multiplication and division needs dedicated instructional time a single day's Engage station cannot supply. Fuchs and Fuchs (2006) and Gersten et al. (2009) describe exactly this combination — grade-level instruction as the default, with a targeted, time-boxed intervention layered alongside it for a specifically diagnosed skill.

What to Look For as an Observer

- Students using the words "decompose," "enclose," and "subtract" — not just describing what they did in their own words.
- During Model It, Two Ways, students checking that both totals match rather than trusting only one method.
- At the T-shape discussion, a student explaining *why* enclose-and-subtract is slower there specifically — because there are two notches, not one — rather than simply stating a preference.
- During Practice: Mixed Shapes, students choosing a method deliberately rather than defaulting to whichever one they tried first.
- Which students need the worked example restated versus which students can start Try It cold — this is the single best predictor of who belongs on the Engage board today, and who may need more than the Engage board can offer (see page 2).
- Exit ticket problems completed with the method labeled, not just a numeric answer — this result, together with today's broader pattern, sets how Day Eight opens.

Appendix: Day Seven in the Grade 6 Math Lesson Plan Template

Template Field	Day Seven Content
Standard(s) & Taxonomy Level	6.G.A.1 (grade level); 3.MD.C.7d (foundational); L.6.4 (ELA/literacy, classwork worksheet); MP.1, MP.7 Apply / Analyze / Evaluate (choosing between two valid strategies)
Learning Target(s) (SWBAT)	SWBAT find the area of a compound figure by decomposing it into rectangles and adding, or by enclosing it in one rectangle and subtracting; SWBAT choose whichever method is more efficient for a given shape.
Success Criteria (I Can)	I can decompose a compound figure into rectangles and add their areas. I can enclose a compound figure in a rectangle and subtract the extra piece. I can choose whichever method is more efficient for a shape. I can see the same shape two different correct ways and get the same area.
Daily Marzano Strategies	Identifying Critical Content; Examining Reasoning (comparing two methods); Practicing Skills, Strategies, and Processes; Cooperative Learning (Journal partner check)
Do-Now (5 min)	Warm-Up Recitation — four add-the-pieces cards, two subtract-the-missing-piece cards, answered chorally.
Instructional Plan	Try It (L-shape, decompose-and-add) → Model It, Two Ways (same L-shape, both methods compared) → Discuss It (T-shape forces a real choice) → Practice: Mixed Shapes (randomized L/T problems) → Engage/Explore/Enrich rotation ends on aligned IXL skills.
Critical Content Debrief (5 min)	Can students state, for a given shape, which method they'd choose and why? Can they explain why a T-shape changes that answer compared to a simple L?
Daily Formative [Ungraded] / Summative [Graded] Assessments	Formative: teacher observation during Model It and Practice (today's diagnostic signal — see page 2), classwork worksheet. Summative: none today; the exit ticket is diagnostic and sets Day Eight's grouping.

Planning to Close the Achievement Gap Using Data

Tier 1 Supports	Tier 2 Supports	Tier 3 Supports
Whole-class Model It, Two Ways walkthrough on the document camera before independent work; grid paper for every problem; both methods named and posted; letter-boxed word bank on the classwork worksheet for every student.	Engage board with the teacher: cut-and-rebuild the L-shape by hand, sentence frames for naming each method, worksheet Part 1 only before attempting Part 2.	Support Room: Multiplication & Division Fluency. Students identified from today's observation (page 2) move to a dedicated support-room block for direct multiplication/division fact and procedure instruction (3.OA.B.6, 3.OA.C.7) — run alongside, not instead of, continued Day 8+ grade-level instruction. Placement revisited weekly against new formative data.

Data Source: Days One–Six formative observations; Day Six Find the Missing Side diagnostic; Day Seven Model It, Practice, and Exit Ticket results; classwork worksheet.

Explore More: Resources for Observers

For guests who would like to look deeper into the curriculum and standards behind today's lesson.

About Today's Curriculum

IXL Skill Alignment, Grade 6

The published Grade 6 skill list under Perimeter and Area (GG), used to choose today's Engage/Explore/Enrich board tasks.

[ixl.com/math/grade-6](https://www.ixl.com/math/grade-6)

Michigan K–12 Math Standards

The Michigan Department of Education's full K–12 Standards for Mathematics document.

[View Michigan DOE standards](#)

Today's Classwork Worksheet — Compound Area

The Guess the Word / Meet the Compound Figure Family worksheet referenced throughout this guide.

[View the worksheet](#)

Standards for Mathematical Practice

The official Common Core description of MP.1, MP.7, and the other six practices.

[View on corestandards.org](https://www.corestandards.org)

References

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